SAVITRIBAI PHULE PUNE UNIVERSITY,

AY 2024-2025

PROJECT REPORT ON ON JOB TRAINING (OJT)

PREPARED BY

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FYMSC(CS)

UNDER THE GUIDANCE OF

ASST.PROF. SONALI GHOLVE

Submitted to Partial Fulfilment of

Msc(Computer Science) Sem-II

SARHAD

SARHAD COLLEGE OF ARTS, COMMERCE AND SCIENCE KATRAJ, PUNE- 411046

Academic Year - 2024-2025

(Minority College)

Sarhad College
of Arts, Commerce & Science (SCACS)

[Affiliated to Pune University (ID No. PU/PN/ACS/300/2008) and Recognised by Government of Maharashtra]

CAAP013640 | A.I.S.H.E. Code : C-41270 | College Code:0847

OJT Project Certificate

This is to certify that ~~Ms~~/Mr. Mali Bhushan Dnyaneshwar satisfactorily completed the OJT Project as required by Savitribai Phule Pune University for M.Sc. (Comp. Sci/Comp. App), Semester –II, in the academic year 2024-2025.

OJT Project Title: Bike Rental System.

Project Guide

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INTERSHIP CERTIFICATE

Praavi Consultants

BHUSHAN
MALI

Praavi Consultants
Pune, 412201
9699369117
01/04/2025

Subject: Internship Offer Letter for Web Developer

Mr. Bhushan Dnyaneshwar Mali
Dear Bhushan Dnyaneshwar Mali,
We are pleased to extend an offer for the position of Web Developer Intern at Praavi Consultants. After reviewing your skills and qualifications, we are confident that you will make a valuable contribution to our team during your internship period.

Position: Web Developer Intern
Joining Date: 01 April 2025
Internship Duration : 3 months.
Working Hours: 9:00 AM - 6:00 PM, Monday to Friday

During your internship, your primary responsibilities will include :

- Assisting in designing, coding, and modifying websites to meet client specifications.
- Collaborating with the team to develop user-friendly interfaces and web applications.
- Debugging and resolving website performance issues.
- Ensuring the websites are optimized for speed, SEO, and responsiveness.
- Learning and implementing the latest web development trends and tools.
- Terms and Conditions
- Your internship is subject to compliance with the company policies and guidelines.
- Confidentiality of company and client information must be maintained during and after the internship.
- The internship may be terminated with a written notice by either party.

We look forward to welcoming you to Praavi Consultants and believe that your skills and experience will contribute significantly to our success. To accept this offer, please sign and return a copy of this letter by 02/04/2024, indicating your agreement with the terms and conditions outlined herein.
If you have any questions or require further clarification, please feel free to contact us at praavi.consultants@gmail.com

Sincerely,
Pooja Badekar
CEO, Praavi Consultants
badekarpooja.praavi@gmail.com

9960943056
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www.praaviconsultants.com

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I am also grateful to my peers and office colleague's for their valuable discussions, moral support in the meetings, and for being a constant source of motivation during this Corporate Project journey.
Last but not least, I owe special thanks to Professors for their unwavering support, patience, and encouragement throughout this endeavor .

Thanking you all.

Mali Bhushan Dnyaneshwar

ABSTRACT

This paper presents the conceptualization, analysis, design, and implementation of a web-based **Bike Rental System** aimed at enhancing urban mobility through a technology-driven rental solution. The system addresses several limitations of traditional rental methods, including manual tracking inefficiencies, lack of real-time bike availability, and suboptimal maintenance workflows. The proposed platform delivers a user-centric interface for seamless bike rentals, while equipping administrators with robust tools for fleet management, booking validation, and operational oversight. Key system modules developed during the internship include a User Module for registration and rental history, a Booking Module enabling real-time reservations and returns, an Admin Module for resource control, and a Bike Management Module for monitoring bike conditions and locations. This project effectively integrates core software engineering principles with real-world transportation challenges, providing not only a functional digital solution but also promoting sustainable and eco-friendly commuting alternatives.

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1. INTRODUCTION

1.1 Company Profile

Praavi Consultants is a rapidly emerging technology company established on July 31, 2023, and headquartered in Pune, Maharashtra, India. company is dedicated to leveraging technology to provide essential tools and expertise, ensuring its clients achieve sustained success in the increasingly digital landscape.

Key Services offered by **Praavi Consultants**:

1. AWS Managed Services
2. DevOps and CloudOps
3. CDN and Security
4. Media Solutions
5. Website Development

1.2 Problem Statement

1: Inefficient Manual Booking Processes:

In many cities, traditional bike rental services still rely on manual logs or token-based systems. This leads to frequent issues such as overbooking, lost rental records, human error, and slow customer service. There's a critical need to automate the booking and rental workflow to improve efficiency and accuracy.

2: Limited Real-Time Visibility:

Customers are unable to check the real-time availability and location of bikes. As a result, users often arrive at empty docking stations or find broken bikes. This results in poor user experience and reduced customer retention.

3: Lack of Maintenance Tracking:

There is no centralized mechanism to report or monitor the condition of bikes. Maintenance is often reactive rather than preventive, causing frequent breakdowns, reduced availability, and safety concerns for users.

4: Fragmented User Experience:

Existing systems often lack mobile responsiveness, smooth navigation, or integrated payment options. Users face challenges in signing up, renting, or returning bikes easily, discouraging widespread adoption of eco-friendly transport options.

5: No Data-Driven Decision-Making:

Bike rental providers lack analytical tools to monitor usage trends, peak hours, and fleet performance. This limits their ability to optimize bike placement, predict demand, or reduce operational costs.

1.3 Existing System & Need of Computerization

Existing bike rentals use manual logs or token-based rentals. Issues include:

- No real-time inventory view
- Difficult to scale
- No automated maintenance tracking

A computer-based system enhances:

- Booking efficiency
- Ride tracking
- Admin control
- Maintenance scheduling

Need for Computerization:

Traditional bike rental services are often dependent on manual processes, such as paper-based records, physical booking counters, and cash transactions. These outdated methods result in errors, delays, poor scalability, and limited data visibility for both users and administrators. As demand for flexible and eco-friendly transport grows, manual systems are no longer sufficient to meet the expectations of modern users who prefer convenience, speed, and real-time access.

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Computerization of the bike rental process introduces a **centralized, automated, and real-time digital platform** that streamlines operations end-to-end. It allows users to register, book, and return bikes through a web or mobile interface anytime, anywhere. Administrators gain access to live system dashboards for monitoring fleet availability, handling bookings, scheduling maintenance, and generating reports.

1.4 Scope of the System

The Bike Rental System is designed to offer a complete digital solution for renting, managing, and maintaining a fleet of bikes in urban and semi-urban areas. The system caters to both end-users and administrators with dedicated modules and functionalities.

For User:

- Register and log in securely through a web interface.
- Browse available bikes based on location or model.
- Book bikes in real time and receive confirmation instantly.
- Unlock bikes using QR code scanning (future enhancement).
- End rides and automatically calculate duration and fare.

1.5 Proposed System

A. User Authentication & Management

- **Login/Signup:** Users can create accounts or log in using their email and password.
- **Profile Management:** Users can update personal details and view booking history.

The platform ensures a user-friendly interface, secure authentication, and scalable backend services to reduce reliance on intermediaries and deliver value to all users.

B. Bike Catalog & Booking System

- **Bike Categories:** Bikes are classified into types (City, Mountain, Electric, etc.) for easy filtering.

- **Real-Time Availability:** Shows which bikes are available for rent at any given time.
- **Dynamic Pricing:** Displays rental rates (e.g., ¥199/hour) with an option for hourly/daily bookings.
- **Instant Booking:** Users can reserve bikes with a single click ("Book Now").

1.6 Objectives of the System

1. Provide an Easy-to-Use Booking System

- Enable users to quickly browse, select, and book bikes with minimal steps.
- Implement an intuitive interface with clear CTAs (e.g., "Book Now").

2. Ensure Secure User Authentication

- Allow users to create accounts and log in securely via email and password.
- Protect sensitive user data with encryption and secure protocols.

3. Offer a Comprehensive Bike Catalog

- Categorize bikes (City, Mountain, Electric, etc.) for easy navigation.
- Display real-time availability, pricing (e.g., ¥199/hour), and descriptions

2. SYSTEM ANALYSIS

2.1 Feasibility Study

1. Technical Feasibility:

- Mature, well-supported tech stack (Angular, Spring Boot, Postgres/MySQL) and cloud infrastructure; team has or can gain required skills.

2. Economic Fesibility:

- Initial investment needed, but long-term cost savings by eliminating intermediaries.

3. Operational Feasibility:

- Simplifies current manual processes for farmers, admins, and customers with userfriendly UI.

2.2 System Requirement Specification (SRS)

2.2.1 Functional Requirements Farmer

Module:

Secure authentication; product listing and management; product request submission.

City Admin Module:

Secure login; dashboard with city metrics; order and request management; profile management.

2.2.2 Non-Functional Requirements Scope:

Performance:

Fast response (2-3 seconds), scalable, 24/7 availability, support for many concurrent users.

Security:

JWT authentication, encryption (in transit & rest), input validation, secure sessions, strict access control.

Usability:

Intuitive, responsive UI (Angular Material), consistent feedback via notifications.

Reliability:

Graceful error handling, regular backups, disaster recovery plan.

Maintainability:

Modular design, coding standards, documentation.

Compliance:

Adherence to data protection laws (e.g., GDPR) and e-commerce best practices.

3. SYSTEM DESIGN

3.1 Design Constraints

Key design constraints that influenced the architecture and implementation of the DoorStep-Agromart project included:

Technology Stack: A predefined technology stack was adopted:

- Frontend:

Angular (with TypeScript, HTML, SCSS, and Angular Material for UI components).

- Backend:

APIs developed using Spring Boot.

- Database:

PostgreSQL or MySQL for relational data storage.

- Performance:

The system design had to prioritize responsiveness and efficient data handling to ensure a smooth user experience, even under varying loads.

- User Interface (UI) / User Experience (UX):

The UI design mandated the use of Angular Material to ensure a consistent, intuitive, and responsive user experience across all modules.

- Security:

Adherence to robust security standards was a critical constraint, necessitating secure authentication (JWT), data encryption, and input validation throughout the system.

- Scalability:

The architecture needed to be designed with future growth in mind, capable of handling an increasing number of users, products, orders, and data volume without significant re-architecture.

- Time and Resources:

Development had to be managed within allocated timeframes and available resources, influencing the scope and prioritization of features in each iteration.

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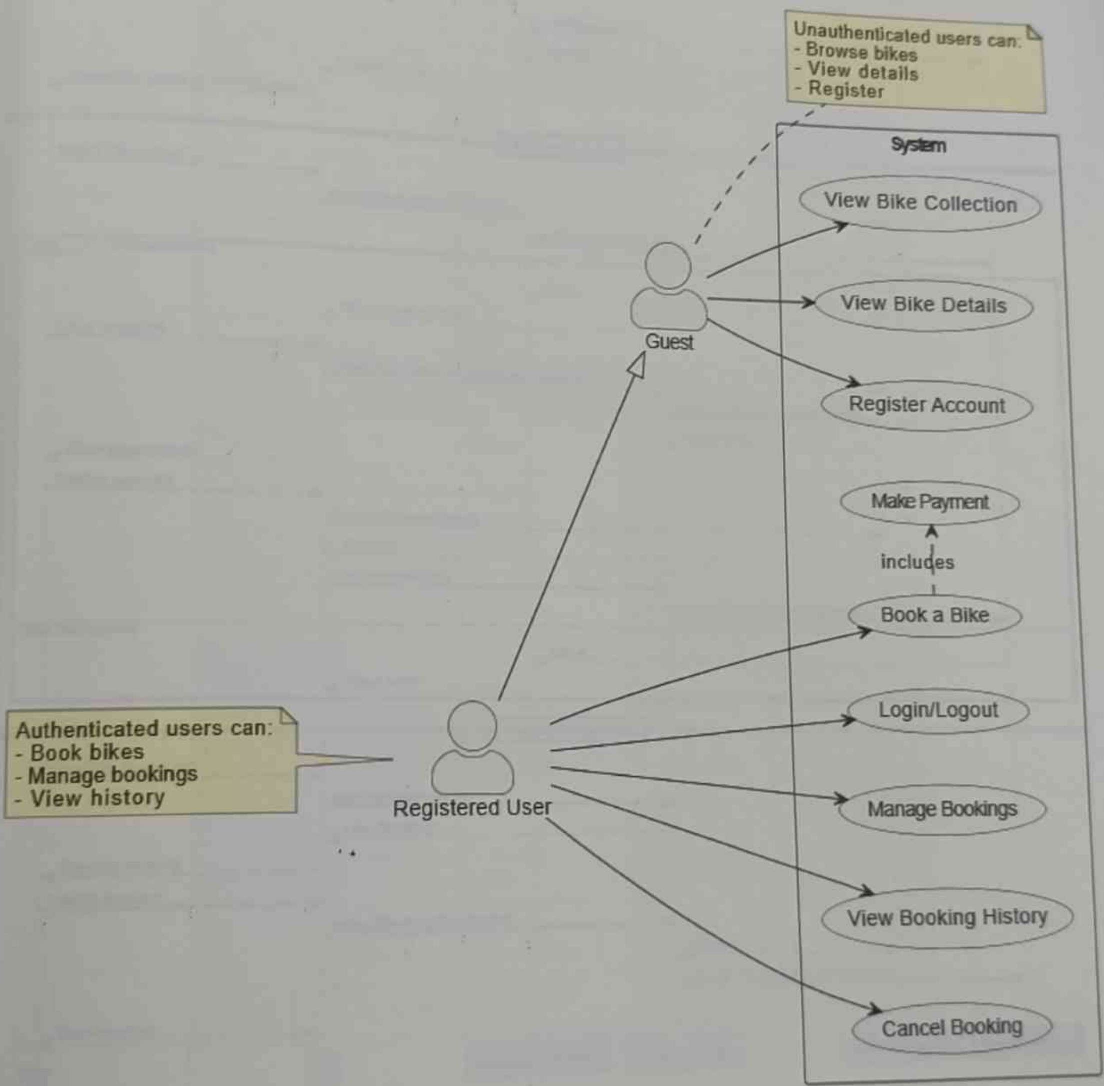
3.2 System Model: UML diagrams

- Use Case Diagram



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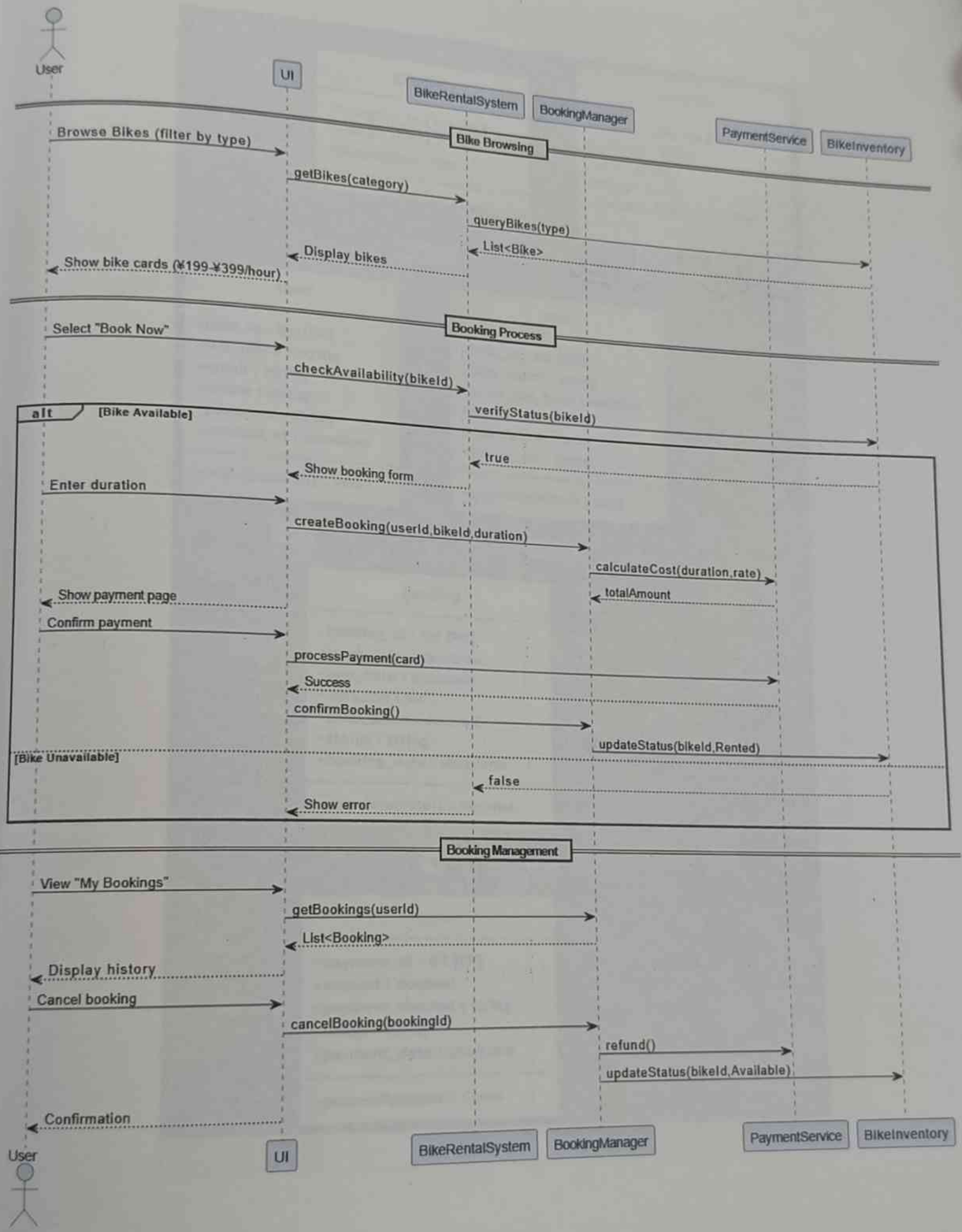
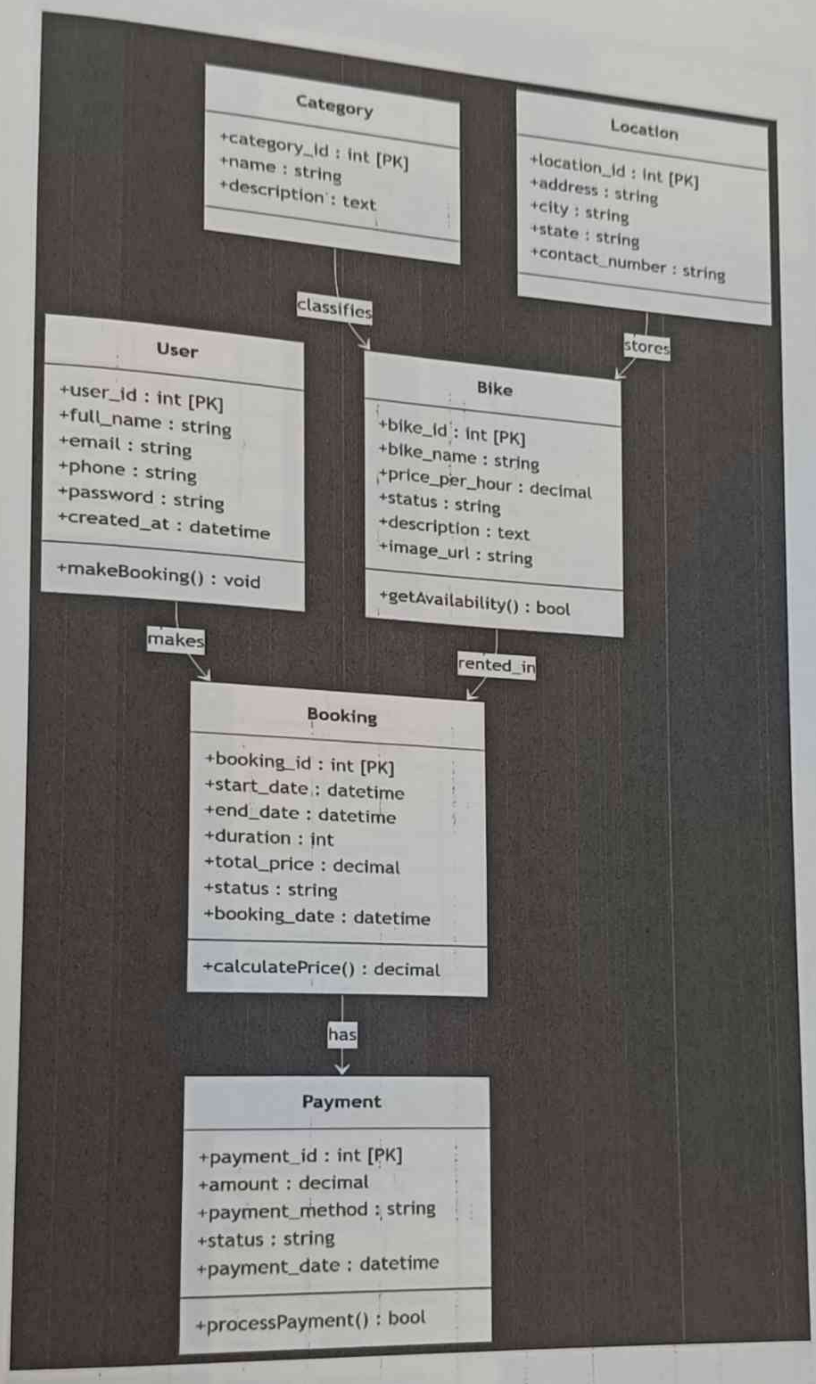
- Sequence Diagram



Class Diagram



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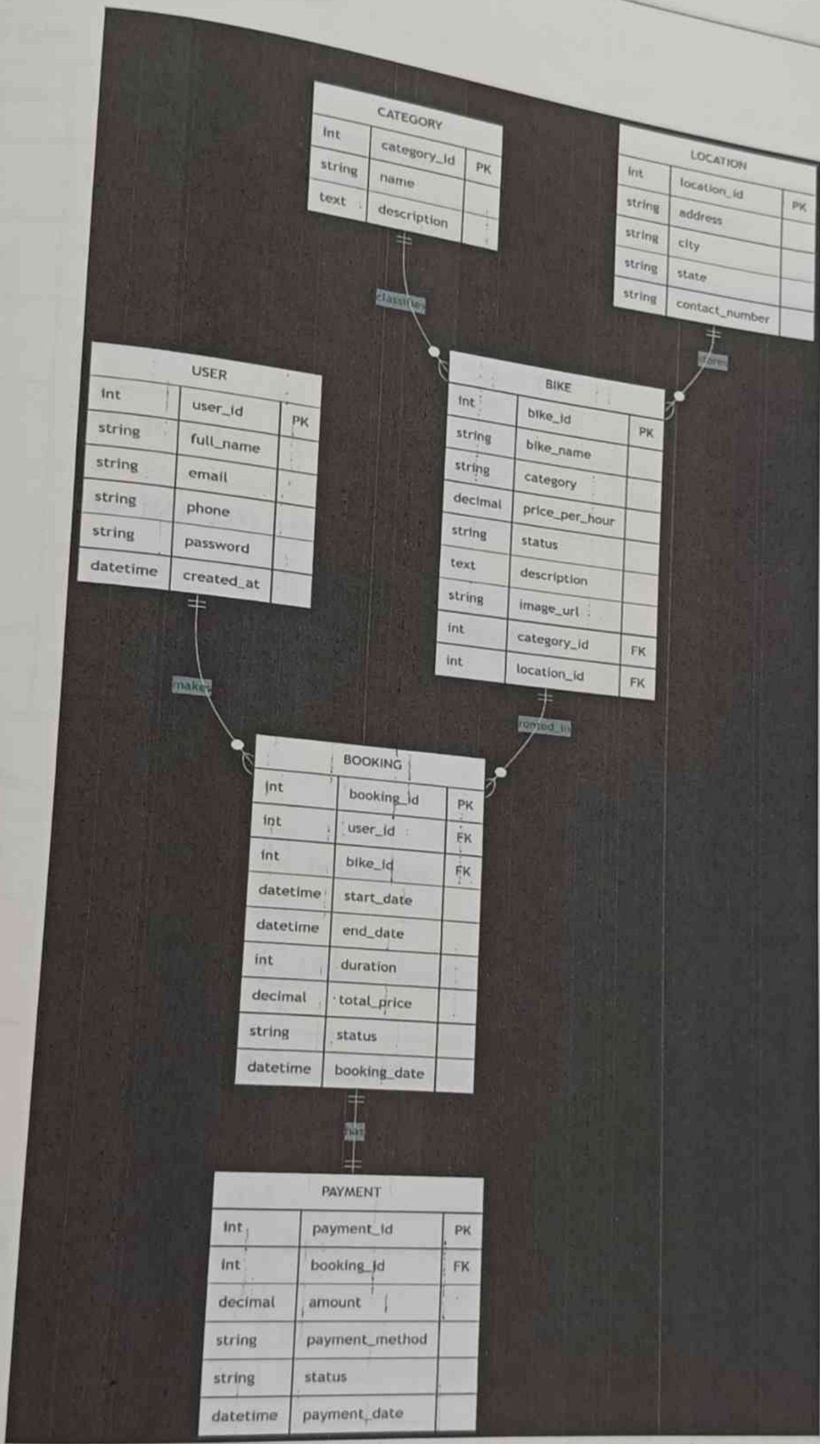
Data Model: ER Diagram



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Data Dictionary Table: Farmer

1. User Table

Field Name	Data Type	Description	Constraints
user_id	INT	Unique identifier for each user	Primary Key, Auto Increment
full_name	VARCHAR(100)	User's full name	Not Null
email	VARCHAR(100)	User's email address	Not Null, Unique
phone	VARCHAR(15)	User's phone number	Not Null, Unique
password	VARCHAR(255)	Hashed password	Not Null
created_at	DATETIME	Account creation timestamp	Not Null, Default Current
status	ENUM	Account status (active/inactive)	Default 'active'

2. Bike Table

Field Name	Data Type	Description	Constraints
bike_id	INT	Unique identifier for each bike	Primary Key, Auto Increment
bike_name	VARCHAR(100)	Name of the bike	Not Null
category_id	INT	Reference to bike category	Foreign Key
price_per_hour	DECIMAL(10,2)	Rental price per hour	Not Null
status	ENUM	Bike status (available/rented/maintenance)	Default 'available'
description	TEXT	Detailed bike description	Null

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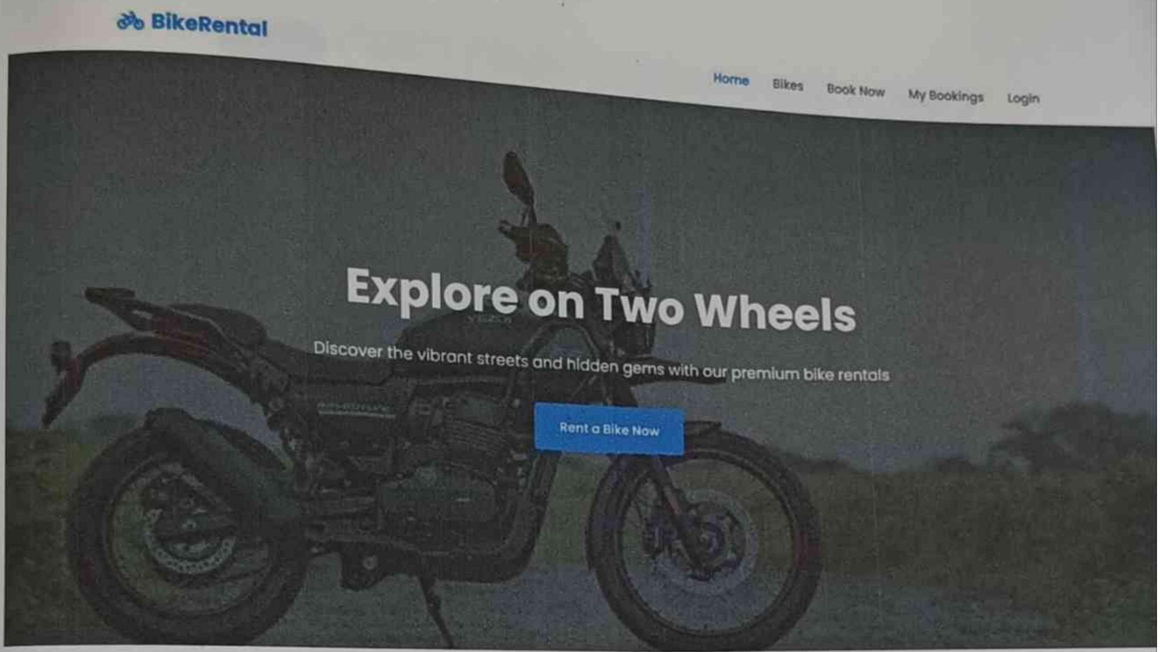
Field Name	Data Type	Description	Constraints
image_url	VARCHAR(255)	URL to bike image	Null
location_id	INT	Current location of the bike	Foreign Key

3. Booking Table

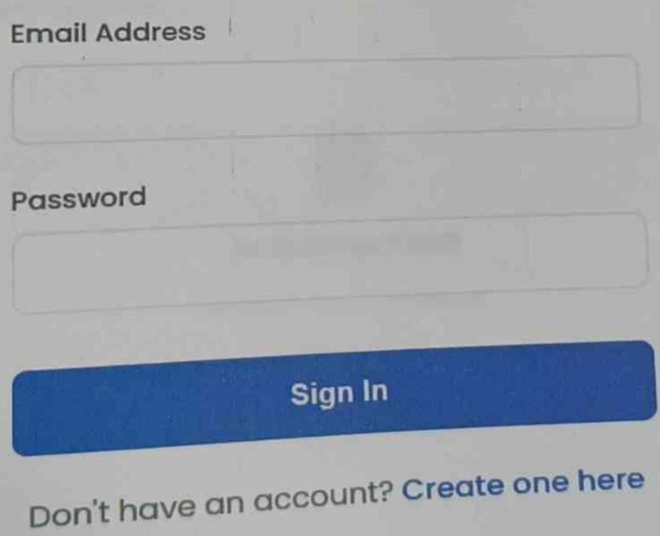
Field Name	Data Type	Description	Constraints
booking_id	INT	Unique identifier for each booking	Primary Key, Auto Increment
user_id	INT	Reference to user making booking	Foreign Key
bike_id	INT	Reference to booked bike	Foreign Key
start_date	DATETIME	Booking start date and time	Not Null
end_date	DATETIME	Booking end date and time	Not Null
duration	INT	Duration in hours	Not Null
total_price	DECIMAL(10,2)	Total booking cost	Not Null
status	ENUM	Booking status (confirmed/cancelled/completed)	Default 'confirmed
booking_date	DATETIME	When booking was made	Not Null, Default Current

3.5. User Interface

- **Home**



Login Form



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Our Bike Collection

All Bikes | City Bikes | Mountain Bikes | Electric Bikes | Road Bikes | Kids Bikes

City Cruiser

City Bike

₹199/hour

Book Now

Mountain Explorer

Mountain Bike

₹299/hour

Book Now

E-Rider Pro

Electric Bike

₹399/hour

Book Now

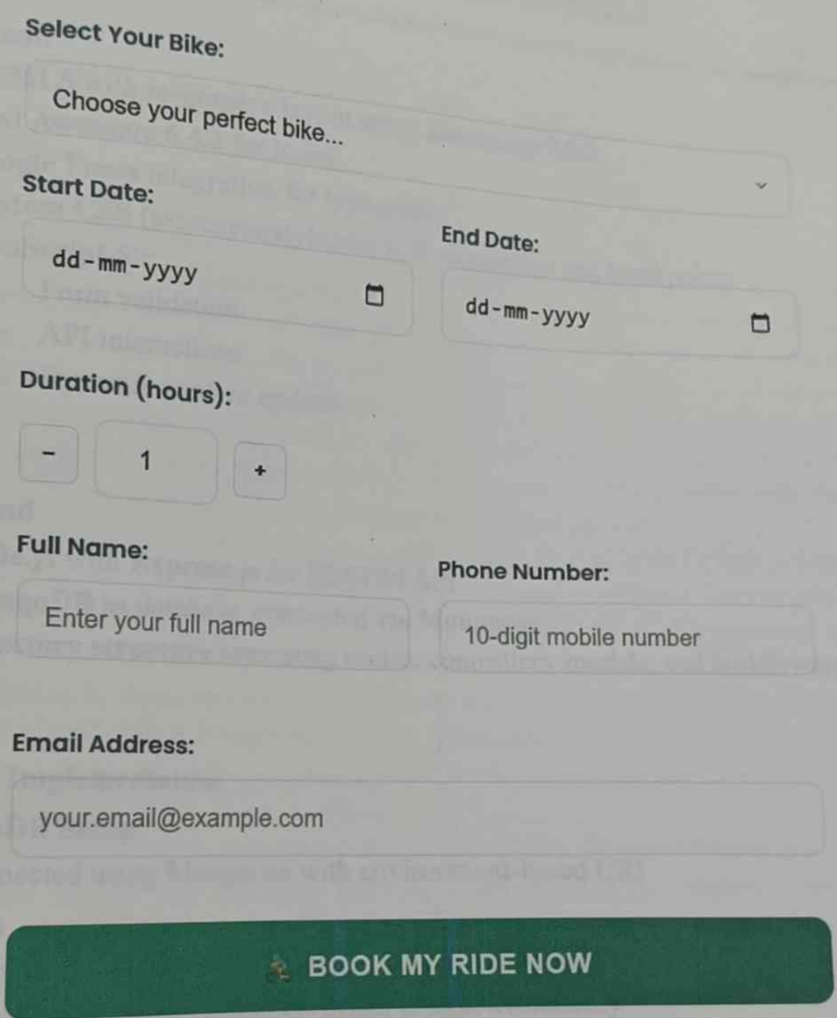
All Bookings | Confirmed | Cancelled

Search bookings...

No Bookings Found

You haven't made any bookings yet.

Rent a Bike


Select Your Bike:
Choose your perfect bike...
Start Date:
dd-mm-yyyy
End Date:
dd-mm-yyyy
Duration (hours):
−
1
+
Full Name:
Enter your full name
Phone Number:
10-digit mobile number
Email Address:
your.email@example.com
BOOK MY RIDE NOW

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4. IMPLEMENTATION DETAILS

◆ **Frontend**

- **HTML5** with responsive layout using **Bootstrap 5.3.2**
- **Font Awesome 6.4.2** for icons
- **Google Fonts** integration for typography
- **Custom CSS** (assets/css/style.css) with animations and breakpoints
- **JavaScript** for:
  - Form validation
  - API interactions
  - Dynamic content updates

◆ **Backend**

- **Node.js** with **Express.js** for RESTful API
- **MongoDB** as database, connected via **Mongoose**
- **Directory structure** separating routes, controllers, models, and middleware

Database Implementation

◆ **MongoDB Setup**

- Connected using Mongoose with environment-based URI

◆ **Models**

- **User Model**: Includes role, email, and hashed password
- **Bike Model**: Name, category, price, image, availability
- **Booking Model**: User ID, bike ID, dates, total price, and status

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5. SYSTEM TESTING

5.1 Test Plan (Descriptive Format)

Testing Plan

The primary objective of this testing plan is to ensure the overall quality, reliability, and usability of the bike rental application by conducting comprehensive functional and non-functional testing. The functional testing will focus on verifying that all features operate according to the defined requirements and user expectations.

In terms of User Authentication, the application will be tested for its login functionality, user registration process, password reset capabilities, and proper session management. This includes ensuring that users can securely log in and out, reset their passwords via email, and maintain authenticated sessions without unauthorized access.

The Bike Management module will be tested to confirm that bike listings are displayed accurately, bike details are accessible and complete, and search and filter functionalities return relevant results based on user inputs. Additionally, it will be validated that bike availability status updates correctly based on booking activity.

For the Booking System, the testing will validate the creation of bookings with appropriate date selection and user information, accurate processing of payment transactions, and the ability to modify or cancel bookings as needed. Each action must reflect accurately in both the system database and the user interface.

The User Dashboard will be reviewed to ensure it correctly displays booking history, allows for seamless profile management, notifies users of relevant updates, and enables submission of reviews for bikes and services. This ensures users can manage their activities and interact with the system effectively.

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Test Case

Test Case ID	Test Category	Description	Input	Expected Result
TC-001	User Authentication	Valid Login	Valid email and password	Successful login, redirect to dashboard
TC-002	User Authentication	Invalid Login	Invalid credentials	Error message displayed
TC-003	User Authentication	Password Reset	Valid email	Reset link sent
TC-004	Bike Management	Bike Search	Search query	Relevant results displayed
TC-005	Bike Management	Filter Bikes	Category/price filter	Filtered results
TC-006	Bike Management	Bike Details	Click on bike	Detailed information displayed
TC-007	Booking System	Create Booking	Bike selection, dates, user info	Booking confirmation
TC-008	Booking System	Payment Processing	Payment details	Payment confirmation
TC-009	Booking System	Cancel Booking	Cancel request	Cancellation confirmation
TC-010	Performance Testing	Concurrent Users	100 concurrent users	Response time < 2 seconds
TC-011	Performance Testing	Database Load	Multiple queries	Query time < 100ms

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Test Case ID	Test Category	Description	Input	Expected Result
TC-012	Security Testing	XSS Prevention	Malicious script	Script sanitized
TC-013	Security Testing	SQL Injection	SQL query in input	Query blocked

6. CONCLUSION

The OJT at **Pravi Consultant** on the **Bike Rental System** project provided practical, hands-on experience in developing a real-time web-based application. During the training, I contributed to the successful development of key modules, including **User Authentication, Bike Management**, and **Booking System**. The project emphasized how digital platforms can streamline rental operations, improve user experience, and support seamless data handling. This experience enhanced my skills in **frontend development, API integration, database interaction**, and **collaborative software development**, while also deepening my understanding of project structure, system analysis, and real-world implementation practices.

7. FUTURE SCOPE

The project has broad potential for growth, including:

Future Scope

1. **Enhanced User Experience**
   The system can be expanded by developing native mobile applications for iOS and Android, offering features like push notifications, offline access, mobile payments, and real-time GPS tracking. Additionally, a Progressive Web App (PWA) can be implemented to enhance accessibility with features like home screen installation, faster load times, and push notifications.
2. **Advanced Features**
   Smart booking systems powered by AI can be introduced to provide intelligent recommendations, predictive availability, dynamic pricing, and smart scheduling. Virtual Reality (VR) integration can further enhance user experience through virtual bike tours, 3D previews, interactive showrooms, and virtual test rides.
3. **Business Expansion**
   The system can scale to support multi-location and franchise-based operations with centralized control, branch-level pricing, and inventory management. For international growth, features like multi-language support, currency conversion, and global payment options can be added.
4. **Additional Services**
   Bike maintenance features such as scheduled servicing, repair tracking, and alerts can be incorporated. Moreover, accessories rental such as helmets, locks, GPS devices, and safety gear can be introduced to enhance service offerings.
5. **Technology Integration**
   Inegration of IoT can bring smart bike features like GPS tracking, app-controlled locking, battery monitoring, and real-time usage statistics. Smart locks with Bluetooth, remote control, and access logging can improve security.

8. BIBLOGRAPHY & REFERENCES

Resources that guided development and design include:

Articles

- Indeed Career Guide, "10 Skills Every Administrative Professional Needs," Indeed.com.
- SHRM, "Best Practices in Office Administration," Society for Human Resource Management (SHRM).

Websites & Online Learning

- W3Schools – Web development tutorials and references. https://www.w3schools.com
- LinkedIn Learning – Courses on Angular, Spring Boot, and software development practices. https://www.linkedin.com/learning
- Coursera – Business and administrative skill-building courses. https://www.coursera.org

Official Documentation

- Angular Official Documentation – https://angular.io/docs
- Spring Boot Documentation – https://docs.spring.io/spring-boot/docs/current/reference/htmlsingle/
- PostgreSQL Documentation – https://www.postgresql.org/docs/
- MySQL Documentation – https://dev.mysql.com/doc/